USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 28 LAB - FMCW Range
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 4.2: I can calculate range resolution, unambiguous range, and Doppler shift for a given radar waveform.

Learning Objective 4.4: I can describe FMCW radar operation and configure an FMCW waveform on the ADALM-PHASER.

Learning Objective 4.5: I can process FMCW radar data to produce range, range-waterfall, and range-Doppler results.

This packet is the L28 Lab, in three parts, and you turn it in once, complete. Do **Part 1** before you go to the bench: the predictions you will test. **Part 2** is your own measurements, from your own kit with the Phaser app, and three short tasks in the course's notebook, run on the captures you saved. Do **Part 3** after the bench: compare the measurements with the predictions and explain the differences. You work this lab on your own, outside class, following the procedure on the L28 page.

Due: see Gradescope.

Grading. The packet is checked and returned with feedback. It is the first piece of scaffolding for your final report, not part of the report: the report explains the radar in your own words, and text lifted from this packet earns nothing there.

The radar is the PHASER's FMCW mode: a 10 GHz carrier, a chirp of bandwidth $B = 500$ MHz and length $T_s = 0.5$ ms, and the Pluto sampling the beat at $f_s = 0.6$ MSPS. Use $c = 3.00 \times 10^8$ m/s. Measure range from the array face.

Part 1: Pre-Lab (individual, before the bench)

1. **From range to beat frequency.** A flat plate stands in front of the kit at the settings above.

   (a) Find the chirp slope, in MHz per microsecond.

   slope = ☐

   (b) Find the round-trip delay of the echo from a plate at 1.00 m, and the beat frequency it produces.

   τ = ☐

   f_b = ☐

   (c) The kit's 0 m frequency is about $f_0 = 100$ kHz. Predict the frequency of the plate's peak at each range.

Range	1.00 m	2.00 m	3.00 m
Beat above f_0			
Peak frequency			

(d) How many cycles of the beat fit in one chirp for the plate at 1.00 m? Show that the number does not depend on T_s.

2. Changing the waveform. Procedure C changes one setting at a time. Predict each row for the plate at 2.00 m, with $f_0 = 100$ kHz.

Settings	**kHz per meter**	**Peak frequency**	**Resolution**	**Reach**
500 MHz, 0.5 ms				
250 MHz, 0.5 ms				
500 MHz, 1 ms				

(a) Show the work for the resolution and the reach of the second row.

(b) The second and third rows put the plate's peak at the same frequency. Name two things the bench can show that tell them apart.

3. The range FFT. One 0.5 ms chirp sampled at 0.6 MSPS.

(a) Find the number of samples in one chirp, the spacing of the FFT's bins in kHz, and that spacing in meters.

N =

Δf =

ΔR =

(b) Two plates stand 0.20 m apart, near 2 m. Does one chirp's FFT show one peak or two? Answer in terms of the beat cycles in one chirp.

(c) Zero-padding the 300 samples to a 4096-point FFT gives bins 0.15 kHz apart. Does it improve the resolution? Explain.

4. Windows and the echo budget.

(a) A strong target and one 30 dB weaker stand 0.9 m apart, three bins. With a rectangular window the strong target's sidelobes there sit about 21 dB below its peak; with a Hann window, about 41 dB. Which window shows the weak target, and what does it give up to do so?

(b) Lesson 25 put a 10 cm corner reflector at 3 m at 31.1 dB above the noise in a 1 MHz bandwidth. One FFT bin is 2 kHz wide. Predict the corner's height above the noise floor in the range profile, rectangular window, and then with Hann.

rectangular =

Hann =

Part 2: Measurements (your kit, and the notebook)

Follow the procedure on the L28 page. Transmit antenna on **OUT2**, connected before you power the kit. Step out of the beam while you read a static target. If a capture fails, use the course's recorded data from the page's *No Hardware?* frame, and mark which rows are recorded rather than your own.

Kit: _____ Target used: _____ Recorded data? _____

5. **Table A: the 0 m frequency** (Procedure A).

Quantity	Measured	Expected
Leakage peak, bench clear		
f_0, plate against the array		
Capture file		

6. **Table B: static ranges** (Procedure B). 500 MHz, 0.5 ms. Compute the last column from your f_0 and 6.67 kHz/m.

Tape range	Peak frequency	Beat above f_0	Range from beat	Capture file
1.00 m				
2.00 m				
3.00 m				

Measured beat per meter, the slope of beat against tape range: | |

Optional, the course corner reflector at 3.00 m: peak _____ dBFS, floor near 10 m _____ dBFS, height _____ dB.

7. **Table C: bandwidth and chirp length** (Procedure C). Plate at 2.00 m. Copy your predictions from Question 2 before you measure.

Settings	Predicted peak	Measured peak	Capture file
500 MHz, 0.5 ms			
250 MHz, 0.5 ms			
500 MHz, 1 ms			

8. **Table D: the waterfall** (Procedure D). Your walk out to about 4 m and back.

Quantity	Measured	Recorded data
Nearest range of your trace		
Farthest range of your trace		
Time for one leg, out or back		
Walking speed from the slope		
Capture file		

9. **Table E: the notebook** (Procedure E). Run `L28_range.ipynb` with the course's `radar_lab.py` on your 2 m capture. Each task changes the one line its cell names. Turn in the saved notebook, with its edits and plots, with this packet.

   Task (a), the window. `WINDOW = "hann"`, then `WINDOW = None`.

Quantity	Hann	Rectangular
Peak range and level		
Strongest sidelobe of the peak		
Floor near 10 m		

   Task (b), zero-padding, on `L28_two_corners.npz`, corners 0.20 m apart. `N_FFT = None`, then `N_FFT = 4096`.

Quantity	No padding	4096 points
Peaks between 1.5 and 2.7 m		
Width 3 dB down		

   Task (c), range from your own f_0. Your `F0` _____ kHz; `f_peak` _____ kHz; `R` _____ against the tape's _____ m. Your changed line:

   _____

Part 3: Writeup (individual, after the bench)

10. **Beat per meter.** Compare the slope from Table B with the 6.67 kHz/m of Question 1. Name the two largest sources of the difference, and say which one moves every range by the same amount and which one changes the slope.

11. **Two ways to halve the beat.** Using Table C and your Question 2 predictions, explain why halving B and doubling T_s give the plate the same beat frequency but not the same range profile.

12. **Reading your waterfall.** Identify every line in your waterfall plot, and say how you found the walking speed. Why can this waterfall give the speed only from the slope of the trace, and not from the phase of the echo?

13. **What you carry forward.** List what you keep from this lab for the L29 Lab and for the final report.

Documentation: _____